

Challenge (Habanero)**Q1.** Solve the system: $x + 2y = 7$, $3x - 2y = 5$. What is x ?

- (A) 2
(B) 3
(C) 4
(D) 5
(E) 6

Q2. A recipe for cake requires $2 \cdot x$ eggs and $3 \cdot x$ cups of flour. If $x + 2y = 12$ and $y = 2$, how many eggs are needed?

- (A) 4
(B) 6
(C) 8
(D) 10
(E) 12

Q3. Solve the system using elimination: $x - 4y = -3$, $2x + 4y = 10$. What is x ?

- (A) -1
(B) 1
(C) 2
(D) 3
(E) 4

Q4. A bridge is 15 meters long. The equation $2x + 5y = 30$ represents the length of the bridge, where x is the length of the road and y is the length of the river. If $x = 5$, how long is the river?

- (A) 2
(B) 4
(C) 5
(D) 6
(E) 8

Q5. The cost of building a house is given by the equations $x + 2y = 15000$ and $3x - 2y = 10000$. What is the cost of materials (x)?

- (A) 2000
(B) 4000
(C) 6000
(D) 8000
(E) 10000

Q6. In a construction project, $3x + 4y = 240$ represents the cost of labor, and $2x - 3y = 60$ represents the cost of materials. What is y ?

- (A) 10
(B) 20
(C) 30
(D) 40
(E) 50

Q7. Solve the system: $2x - 3y = 5$, $4x + 5y = 17$. What is y ?

- (A) -2
(B) -1
(C) 1
(D) 2
(E) 3

Q8. A bakery sells a total of 250 loaves of bread and pastries per day. The equation $x + y = 250$ represents the total sales. If $2x - 3y = 50$, how many pastries (y) are sold?

- (A) 50
(B) 75
(C) 100
(D) 125
(E) 150

Open Ended Questions (FRQ)**Q9.** Solve the system of equations: $x + 2y - z = 7$, $2x - 3y + 4z = -2$, $x + y + 2z = 4$. Show all work and find the value of x , y , and z . If the system has no solution or infinitely many solutions, explain why.**Q10.** A contractor is building a new house. The cost of materials is given by $2x + 5y = 15000$, where x is the cost of lumber and y is the cost of cement. The cost of labor is given by $x - 2y = -3000$. Find the cost of lumber (x) and the cost of cement (y). Explain your method and show all work.**Chef's Tips**

- Emphasize the importance of checking solutions in the original equations.
- Encourage students to use substitution and elimination methods.
- Remind students to consider the constraints of the problem, such as non-negative values for lengths and costs.